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A cross-domain few-shot remaining useful life estimation framework based on model-agnostic meta-learning with task embeddings

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ABSTRACT

Remaining useful life (RUL) estimation aims to predict the time until system failure based on monitoring data, facilitating proactive maintenance actions. Precise RUL estimation can significantly enhance system reliability and safety. However, when new system failures emerge, predictive models trained on historical failures data often encounter difficulties in accurate estimation. The distribution shift between historical and new failures data, coupled with extremely few new failures data, results in cross-domain few-shot prognostic scenarios, posing a significant challenge to many deep-learning-based RUL estimation methods. In response to the challenge, this paper proposes a novel cross-domain few-shot RUL estimation framework based on model-agnostic meta-learning (MAML) with task embeddings. First, a segmentation strategy is adopted to construct more meta-tasks, which can capture more comprehensive degradation information for efficient meta knowledge extraction. Then, task embeddings that are independent of backbone network are designed to encode task-specific degradation knowledge into efficient low-dimensional vectors, which alleviates overfitting caused by limited labeled data, thus improving RUL estimation performance. Moreover, the encoded degradation knowledge is only injected into feature extractor, making representation change dominant for better cross-domain adaptability. Experimental results on turbofan engine and wind turbine gearbox datasets reveal the effectiveness and superiority of the proposed framework. Estimation results evaluated by RMSE and Score improve 9 % and 31 %, respectively.

1. Introduction

Prognostics and health management (PHM) technology aims to accurately predict the health state of mechanical systems and perform reasonable maintenance and management based on the health state, which can effectively ensure the safety, reliability and economy of system operation (Márquez et al., 2023; Li et al., 2023; Souza et al., 2023). As a key component of PHM (Camci, Chinnam, 2010), remaining useful life (RUL) estimation involves the analysis of monitoring data or the formulation of degradation models to predict machinery RUL (Soualhi et al., 2023). The accurate RUL estimation of machinery allows the timely improvement of maintenance plans to ensure the smooth progress of industrial activities, as well as reduce maintenance costs, thereby significantly improving the reliability and safety of machinery (Schwartz et al., 2022). Recently, deep learning methods such as convolution neural networks (CNNs) (Li et al., 2018; Al-Dulaimi et al., 2019), long short-term memory (LSTM) (Zhang et al., 2019; Calvo-Bascones and Sanz-Bobi, 2023), gated recurrent units (GRUs) (Luo et al., 2020), neural turing machines (NTMs) (Falcon et al., 2022), attention mechanism (Xu et al., 2022; Du et al., 2023), and variational auto-encoders (Ping et al., 2019) have achieved satisfactory performance for the RUL estimation of mechanical systems. However, in most previous deep-learning-based methods, historical (base) and current (novel) domain monitoring data are assumed to have similar distributions, which is inconsistent with cross-domain PHM applications (Costa et al., 2020). Distribution shift always exists across historical and current domains because of different operational conditions, fault modes, and system noise.

In response to the above cross-domain scenarios, transfer learning (TL) has been employed as an effective way of transferring the knowledge learned in the historical domain to the current domain. It focuses on building models that can fulfil the tasks of the current domain by learning knowledge from a semantically related but differently distributed historical domain (Zhang and Gao, 2024). Fu et al. (2021)

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simultaneously integrated the multi-kernel maximum mean discrepancy and adversarial mechanisms to reduce the domain discrepancy for better cross-domain performance. Lu et al. (2025) designed a pseudo-label-guided domain adversarial network with a dynamic penalty term to dynamically adjust the conditional distribution during joint distribution alignment. Nejjar et al. (2024) aligned the operation profile to execute domain adaptation for better RUL estimation. Most of the abovementioned methods belong to the category of domain adaptation or transductive TL (Pan and Yang, 2010), and are often used in scenarios where available data in the current domain is sufficient and unlabeled (Fan et al., 2020). However, when the data from the current domain is insufficient and labeled, that is, cross-domain few-shot RUL estimation scenarios, the performance of these transductive TL methods might be limited as they do not utilize any supervised information from the current domain.

In contrast, inductive TL (Pan and Yang, 2010) methods are more effective in cross-domain few-shot scenarios. They can utilize the supervised degradation information from some of the labeled data in the current domain to induce an objective predictive model. Pretrain-finetune is a typical inductive TL strategy. Zhang et al. (2018) and Gribbestad et al. (2021) employed LSTM networks to construct pretrain-finetune-based frameworks for cross-domain few-shot RUL estimation. Although pretrain-finetune approaches have achieved good results in some easy RUL estimation scenarios, performance limitation or even negative transfer may occur in cases of considerable domain divergences (Yang et al., 2022). Meta-learning is another promising way of adopting the inductive TL setting to address the cross-domain few-shot RUL estimation issue, which enables model generalization from limited data by learning the meta knowledge shared among tasks. Model-agnostic meta-learning (MAML) (Finn et al., 2017) has attracted attention within the field of RUL estimation in recent years as a general meta-learning algorithm. Ding et al. (2021) leveraged MAML to develop multiple meta-learning prognostic structures based on CNN and GRU. Mo et al. (2023) constructed pseudo-meta-RUL tasks and employed MAML for the RUL estimation of aircraft engines. However, there are still some limitations in these cross-domain few-shot RUL estimation frameworks using the MAML.

(1) These frameworks typically construct the support set of a meta-task using trajectories from the base domain. However, in industrial scenarios, the insufficient number of available trajectories can hinder efficient meta-knowledge extraction during meta-training, potentially reducing model accuracy.

(2) These frameworks update the entire model using few labeled samples to learn task-specific degradation knowledge in the adaptation process. However, computing gradients in a high-dimensional parameter space using limited samples may cause overfitting.

(3) These frameworks rely on representation reuse for adaptation. However, its success depends on the similarity between base and novel domains, and the restricted change in the feature space of degradation representations can limit model adaptability in cross-domain scenarios.

To tackle the aforementioned challenges, a cross-domain few-shot RUL estimation framework based on an improved MAML is proposed in this study. The major contributions of this paper are summarized as follows.

- A segmentation strategy is specifically designed for constructing meta-tasks by dividing a single trajectory into several sub-trajectories. The innovative strategy significantly increases the number of meta-tasks, enabling meta-learning algorithms to derive more generalized meta knowledge from the expanded meta-tasks.
- Task embeddings are developed to refine the MAML algorithm for RUL estimation scenarios. Instead of updating the entire predictive model parameters, only the task embeddings are updated based on a few labeled samples in the adaptation process. This targeted update allows for the acquisition of task-specific degradation knowledge by computing gradients in a low-dimensional parameter space, effectively reducing the risk of overfitting in scenarios with limited labeled samples.
- Representation change is employed as the main adaptation strategy instead of representation reuse by exclusively linking task embeddings with the feature extractor. In this new way, the feature space of the extracted degradation representations can be significantly changed to adapt to new RUL estimation tasks, thereby significantly enhancing the model cross-domain adaptability.

The remainder of this paper is structured as follows. Section 2 outlines the relevant theoretical backgrounds. Section 3 describes the proposed RUL estimation framework in detail. In Sections 4, 5, 6, and 7, the experimental results are analyzed from various perspectives to determine the superiority and effectiveness of the proposed framework. Finally, Sections 8 and 9 respectively present the discussions and the conclusions.

2. Theoretical backgrounds

2.1. Meta-learning

Meta-learning aims to make models "learning to learn" by utilizing episodic training paradigm (Wang et al., 2020; Ma et al., 2022). The objective is to learn a learning strategy, often termed as meta knowledge, from the base domain, enabling models to adapt to tasks with a few labeled samples in the novel domain. To achieve the learning strategy, a series of meta-tasks in the base domain are constructed to simulate the testing circumstance in the novel domain where only a handful of labeled samples are available. The distinct mechanism of meta-learning empower the model with powerful generalization capability in the novel domain (Feng et al., 2022). Each base domain meta-task and each novel domain task are both composed of a support set and a query set. A series of base domain meta-tasks constitutes the meta-training set $D^{train} = \{T_i^{train}\}_{i=1}^{Num} = \left\{ \left(D_{s,i}^{train}, D_{q,i}^{train} \right) \right\}_{i=1}^{Num}$ and a series of novel domain tasks constitutes the meta-testing set $D^{test} = \left\{T_j^{test}\right\}_{j=1}^{num} = \left\{ \left(D_{s,j}^{test}, D_{q,j}^{test} \right) \right\}_{j=1}^{num}$, where Num refers to the number of base domain meta-tasks, num refers to the number of novel domain tasks, s represent the support set and q represent the query set.

Meta-learning algorithms generally are performed via two types of learners. Meta learner learns meta knowledge across a distribution of tasks such as a metric space, a good initialization, and task representations. Base learner utilizes the meta knowledge to quickly learn task-specific knowledge with limited labeled samples (Baik et al., 2020b).

2.2. Model-agnostic meta-learning

Model-agnostic meta-learning (MAML) is a gradient-based meta-learning algorithm proposed by Finn et al. (2017), where the meta learner obtains a common initialization among tasks by gradient descent. It is called model agnostic because it can be used with different network structures and loss functions.

MAML performs optimization in two phases: meta-training phase and meta-testing phase as shown in Fig. 1. Meta-training learns a good initialization from the base domain, enabling meta-testing to accomplish rapid adaptation to new tasks with a few labeled samples from the novel domain. In the meta-training phase, MAML is updated with an interleaved training procedure comprised of the inner and outer loop, which is implemented on a batch of meta-tasks $\{T_i^{train}\}_{i=1}^{M} = \left\{ \left(D_{s,i}^{train}, D_{q,i}^{train} \right) \right\}_{i=1}^{M}$ at each iteration, where M is the meta batch-size. In the inner loop, each base learner is optimized by one or more gradient update steps on the support set $D_{s,i}^{train}$ to learn task-specific knowledge as:

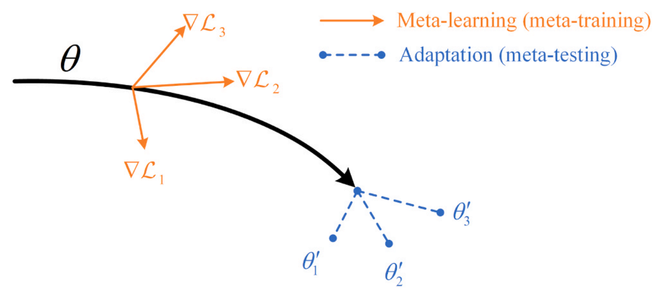

Fig. 1. The schematic diagram of MAML.

$$\theta_i' \leftarrow \theta - \alpha_{inner} \cdot \nabla_\theta \mathscr{L}(\theta, D_{s,i}^{train}), \tag{1}$$

where θ is the initial parameters, θ_i' is the updated parameters corresponding to the base learner of meta-task T_i^{train} and α_{inner} is the learning rate of the inner loop. In the outer loop, the meta learner synthesizes multiple losses of the updated base learners and applies one meta-gradient update step via (2) to obtain a good initialization:

$$\theta \leftarrow \theta - \alpha_{outer} \cdot \nabla_\theta \frac{1}{M} \sum_{i=1}^{M} \mathscr{L}(\theta_i', D_{q,i}^{train}), \tag{2}$$

where α_{outer} is the learning rate of the outer loop.

In the meta-testing phase, the adaptation process and the evaluation process are performed. For the jth new task, the model accomplishes adaptation by one or more gradient update steps on the support set $D_{s,j}^{test}$ based on the optimized initialization θ:

$$\theta_j' \leftarrow \theta - \alpha_{inner} \cdot \nabla_\theta \mathscr{L}(\theta, D_{s,j}^{test}), \tag{3}$$

Subsequently, model evaluation is conducted on the query sets $D_{q,j}^{test}$.

In summary, MAML aims to find an effective initialization that may not achieve optimal performance in the tasks during the meta-training phase, but facilitates rapid adaptation to new tasks during the meta-testing phase.

3. Proposed framework for RUL estimation

3.1. Problem description

RUL estimation of mechanical systems is a regression problem. The features extracted from the multi-sensor monitoring data are used to form the inputs $X_d \triangleq \{x_{d,1}, x_{d,2}, \ldots, x_{d,t_d}\}$ and the corresponding RUL values are used to form the outputs, where t_d means the number of recorded time points in the dth mechanical system. The model needs to learn the mapping relationship between the inputs and the outputs by training data and then is applied for real-time monitoring data of mechanical systems to predict accurate future failure time.

In prognostic applications, particularly when dealing with complex mechanical systems, practitioners frequently encounter cross-domain few-shot scenarios. A chronological overview of cross-domain few-shot prognostic scenarios is depicted in Fig. 2. The predictive models $F_B(\cdot)$ are first developed in Stage 1 using historical data with available labels (D^{train} from the base domain). During Stage 2, the equipment is deployed to the application. In this stage, new faults could emerge and unlabeled data can be observed. Subsequently, the transition to Stage 3 occurs with the first batch occurrence of failures (Fan et al., 2020), where few labeled data with new faults (D_s^{test} from the novel domain) are available, i.e., few-shot. Based on the degradation information from $F_B(\cdot)$, these few labeled data are employed to induce new predictive models $F_N(\cdot)$ for better adaptation to new faults (i.e., cross-domain few-shot scenarios).

To provide further clarification, the following are the symbols definitions for the problem description. In cross-domain few-shot scenarios, labeled samples are extremely few (only K) when novel domain emerges. Therefore, the supervised information in the base domain with relatively sufficient data should be extracted to enhance the model generalization ability in the novel domain. Because RUL labels of all time points in one trajectory are usually obtained concurrently after failure, it is more practical to treat each run-to-failure trajectory as one few-shot unit. The base domain training data is denoted as $D^{train} = \left\{\left(x_{d,1}^{train}, RUL_{d,1}^{train}\right), \left(x_{d,2}^{train}, RUL_{d,2}^{train}\right), \ldots, \left(x_{d,t_d}^{train}, RUL_{d,t_d}^{train}\right)\right\}_{d=1}^{T}$, where T represents the total number of run-to-failure trajectories in the base domain. For the jth task in the novel domain, it is assumed that there exists a limited number of run-to-failure trajectories represented as $D_{s,j}^{test} = \left\{\left(x_{d,1}^{test}, RUL_{d,1}^{test}\right), \left(x_{d,2}^{test}, RUL_{d,2}^{test}\right), \ldots, \left(x_{d,t_d}^{test}, RUL_{d,t_d}^{test}\right)\right\}_{d=1}^{K}$, where K represents the total number of run-to-failure trajectories in the novel domain. The problem is to predict the RUL values of recently produced or existing truncated trajectories in the novel domain by leveraging the information gleaned from the base domain trajectories alongside the K novel domain trajectories. Regarding the issue, after capturing meta degradation knowledge in the base domain via meta-learning, task adaptation is accomplished by only K run-to-failure trajectories in the novel domain and the RUL of truncated test trajectories are predicted.

3.2. Prognostic scheme

The cross-domain few-shot RUL estimation framework developed in this paper is composed of two main stages: the meta-tasks construction and meta-learning stages, as illustrated in Fig. 3. In the first stage, a series of meta-tasks is constructed and enriched by segmentation. In the second stage, MAML with task embeddings is employed to perform meta-training and meta-testing, where the task embeddings are designed to mitigate overfitting, and representation change is adopted as the main adaptation strategy to improve cross-domain adaptability.

3.2.1. Meta-tasks construction

To construct a series of meta-tasks that can simulate the testing circumstance in the novel domain, T trajectories in the base domain are

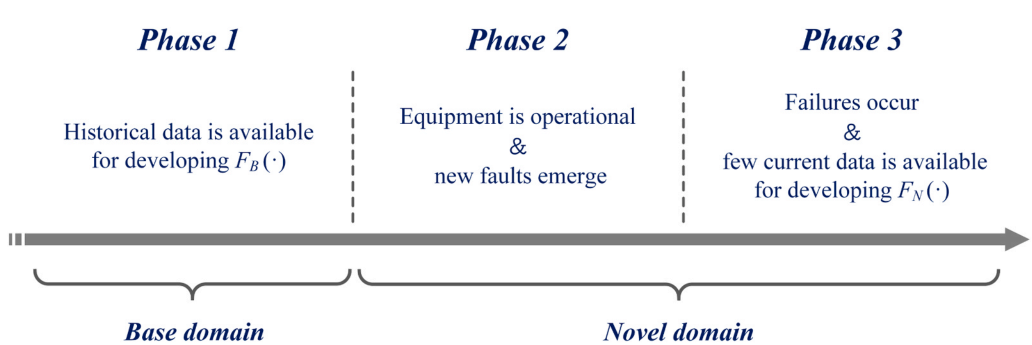

Fig. 2. The chronological overview of cross-domain few-shot prognostic scenarios.

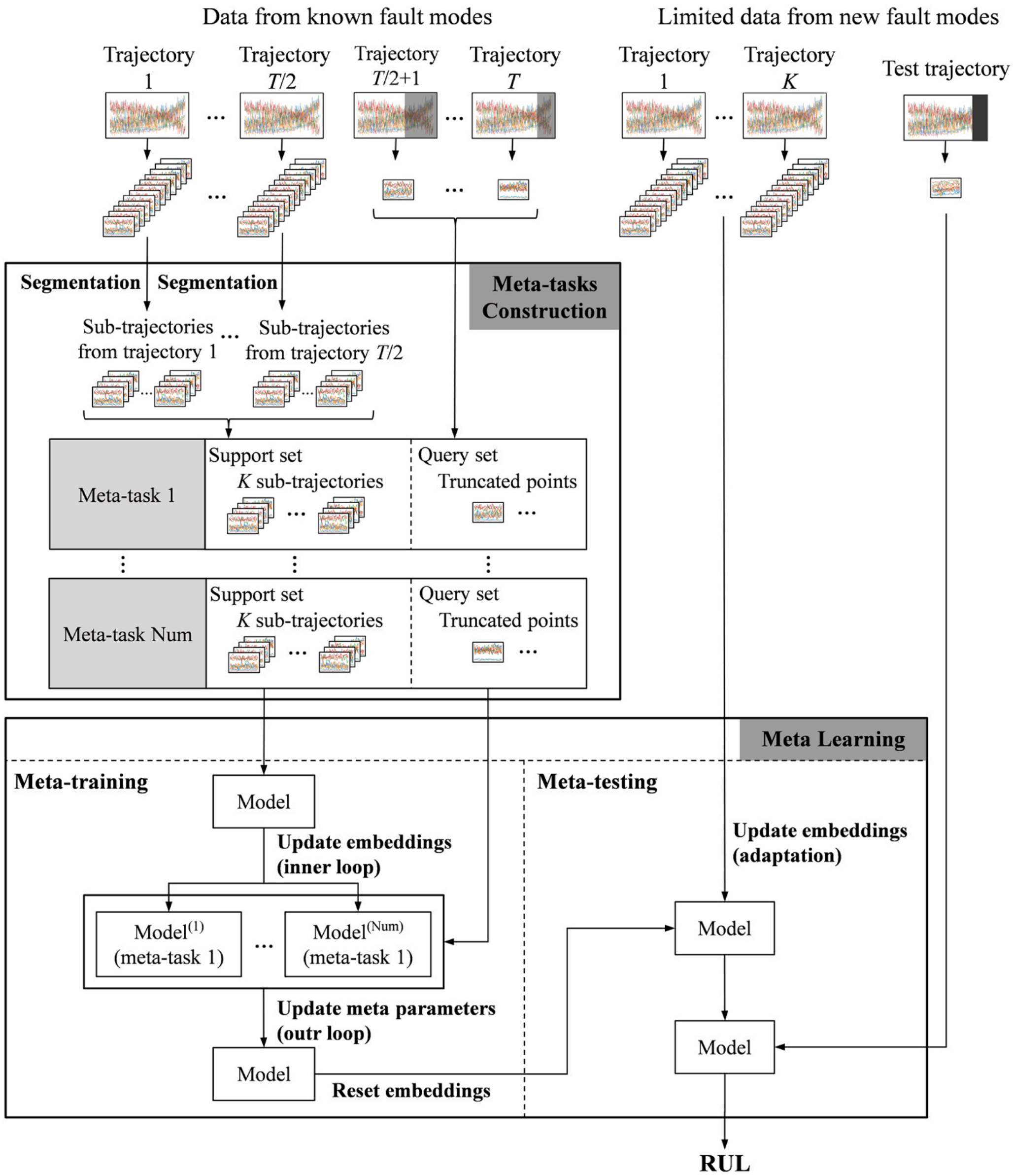

Fig. 3. The flowchart of the proposed framework.

divided equally into two parts, where each part contains $T/2$ trajectories. K trajectories are then randomly and non-repetitively sampled from the first part as the support set of a meta-task. Various truncation points are randomly and non-repetitively selected in the trajectories from the second part as the query set of the meta-task. A series of meta-tasks is constructed by repeating the above operation. However, if there is an inadequate number of trajectories from the base domain, the constructed meta-tasks may be not enough for efficient learning by the meta-learning algorithm to learn good meta knowledge.

In view of the above situation, a segmentation strategy is adopted for meta-tasks construction. One run-to-failure trajectory is divided equally into P segments and one point is sampled randomly and non-repetitively in each segment. These sampled points are then grouped together to form a sub-trajectory. Through segmentation, multiple sub-trajectories are obtained from one trajectory and K sub-trajectories are randomly selected to construct the support set of the meta-task, which increases the number of meta-tasks and thus offers more degradation information for meta knowledge extraction during meta-training. Notably, the proposed segmentation strategy, while theoretically similar to random sampling techniques used to generate varied training samples, is distinct in both its function and application within meta-learning-based RUL estimation framework. In contrast to typical random sampling strategies that often divide data without preserving its chronological integrity, the proposed method strategically divides degradation trajectories to ensure that each sub-trajectory consistently represents a complete phase of the degradation process. This is crucial for maintaining the integrity of each stage within the sub-trajectories, providing a solid foundation for effective meta-training.

3.2.2. Model-agnostic meta-learning with task embeddings

The MAML-based RUL estimation frameworks always update all the predictive model parameters in the adaptation process using a handful of data. In this way, task-specific knowledge is encoded in a high-dimensional parameter space using only limited samples, which leads to overfitting (Mishra et al., 2018). In contrast, encoding task-specific knowledge in an additional low-dimensional parameter space can effectively alleviate overfitting and improve the generalization of predictive models when the labeled data is limited (Rusu et al., 2019; Zintgraf et al., 2019; Flennerhag et al., 2020). Based on this theory, the task embeddings in our algorithm, kind of additional parameters, are designed to improve the few-shot RUL estimation performance. Task embeddings are efficient low-dimensional vectors that are optimized in the inner loop and the adaptation process to learn task-specific degradation knowledge. As shown in Fig. 4, task embeddings φ and shared meta-parameters θ constitute the entire predictive model, where task embeddings φ infuse encoded task-specific degradation knowledge into the meta-parameters θ via feature-wise linear modulation (FiLM) (Perez et al., 2018), as shown in Fig. 5. FiLM affects the output of the network by adaptively performing a simple, feature-wise affine transformation of the intermediate features. Consequently, the embeddings can efficiently manipulate the intermediate features of the meta-parameters in a selective and meaningful manner using FiLM layers. Specifically, considering a convolutional layer output feature map $\{f_l\}_{l=1}^{L}$ with L channels, the output of the lth channel after being connected to the embeddings can be expressed as: $FiLM(f_l) = \gamma_l f_l + \beta_l$, where $\gamma, \beta \in \mathrm{R}^L$ are the transformations of the embeddings. The fully-connected layers, which are denoted as transformation layers in Fig. 4, are employed to implement the transformation. The first half of the output represents γ and the second half of the output represents β. In addition, it should be noted that the proposed algorithm remains model agnostic, which means that the backbone shown in Fig. 4 has the flexibility to be substituted with alternative model architectures.

In the inner loop and adaptation process, φ are updated by the base learner in one or more gradient update steps to learn task-specific degradation knowledge as:

$$\varphi_i \leftarrow \varphi - \alpha_{inner} \cdot \nabla_{\varphi} \mathscr{L}(\theta, \varphi, D_{s,i}^{t}), \tag{4}$$

where $t = train$ implies in the inner loop and $t = test$ implies in the adaptation process. Additionally, $\mathscr{L}$ represents the mean squared error (MSE) loss between the estimated RUL and actual RUL. In the outer loop, θ is updated explicitly by the meta learner in one meta-gradient update step to capture meta degradation knowledge as:

$$\theta \leftarrow \theta - \alpha_{outer} \cdot \nabla_{\theta} \frac{1}{M} \sum_{i=1}^{M} \mathscr{L}(\theta, \varphi_i, D_{q,i}^{train}). \tag{5}$$

Task embeddings are comprised of a set of low-dimensional vectors; thus, it can be taken as a low-dimensional parameter space. In contrast, the entire predictive model consists of multiple convolution and fully connected layers, making it a high-dimensional parameter space. Unlike MAML, which encodes task-specific degradation knowledge into a high-dimensional parameter space by updating the entire predictive model in the adaptation process, the proposed algorithm encodes task-specific degradation knowledge into a low-dimensional parameter space by only updating task embeddings during adaptation. Consequently, the adaptation process, where labeled data is limited, can be decoupled from the high-dimensional parameter space, which effectively attenuates overfitting by avoiding operations that employ very few labeled data to train high-dimensional parameter spaces. The implementation of the improved MAML is outlined in Algorithm 1.

Task embeddings can be connected to the feature extractor or the regressor to constitute a complete predictive model through FiLM. However, if task embeddings are only connected to the regressor, or connected to both the feature extractor and the regressor, the cross-domain ability of the predictive model becomes limited because the algorithm tends to optimize the decision boundary of the regressor rather than the degradation representation space of the feature extractor in the adaptation process, that is, representation reuse (Raghu et al., 2020) becomes the dominant adaptation strategy. In this case, the algorithm lacks the ability to make a significant change to the degradation representation space, hindering the adaptation process of the predictive model for new tasks. In our algorithm, task embeddings are only connected to the feature extractor, as shown in Fig. 4 and Fig. 5. Therefore, the predictive model can exclusively optimize the degradation representation space of the feature extractor in the adaptation process, making representation change (Oh et al., 2021) the dominant adaptation

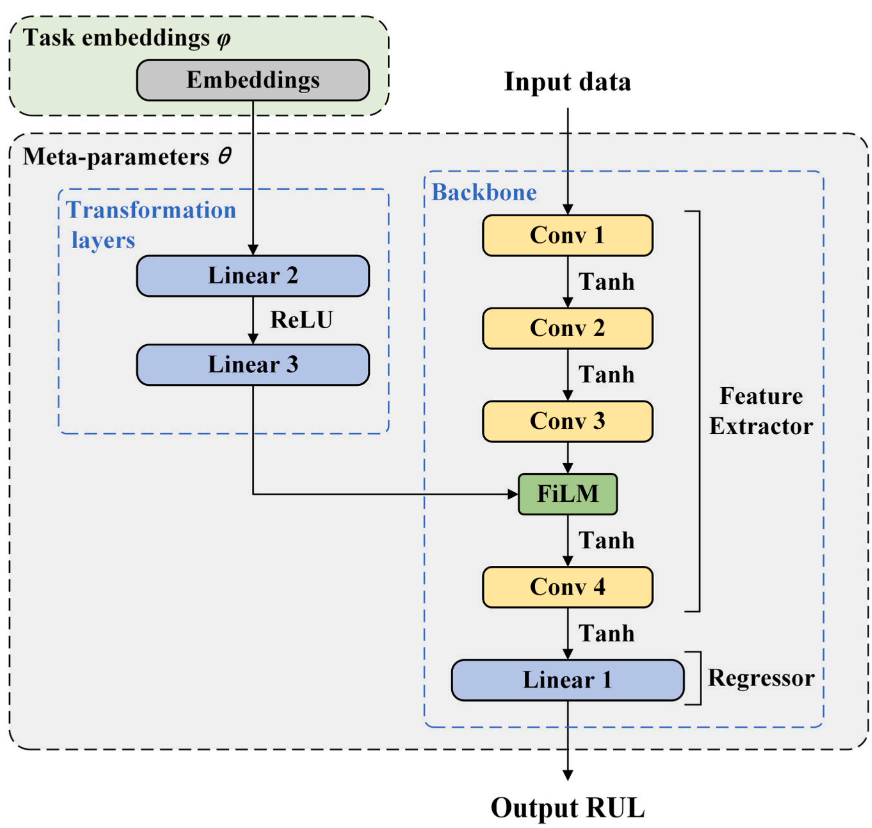

Fig. 4. The overview of the predictive model. Meta-parameters encode meta degradation knowledge and task embeddings encode task-specific degradation knowledge.

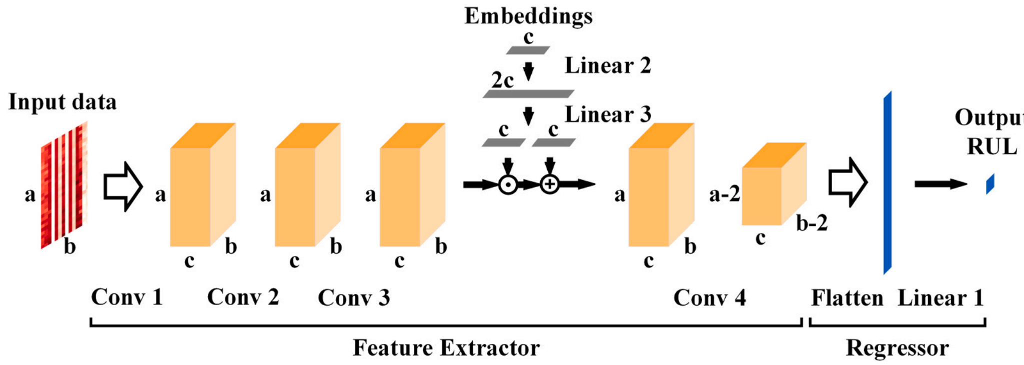

Fig. 5. The details of the embeddings. The embeddings inject task-specific degradation knowledge to the feature extractor by feature-wise linear modulation. The variable "a" represents the time dimension size, the variable "b" represents the sensor dimension size, and the variable "c" represents the channel size.

strategy. In this case, the representation space can be significantly changed to adapt to the new tasks by accepting task-specific knowledge from task embeddings, which improves cross-domain adaptability.

Algorithm 1. : MAML with task embeddings for cross-domain few-shot RUL estimation

Table 1
Details of C-MAPSS dataset.

Sub-dataset	FD001	FD002	FD003	FD004
Time series training set	100	260	100	249
Time series test set	100	259	100	248

Require: D^{train}: meta-training set; T_i^{train}: i th meta-task from meta-training set; $D_{s,i}^{train}$: the support set of T_i^{train}; $D_{q,i}^{train}$: the query set of T_i^{train}; D^{test}: meta-testing set; T_j^{test}: jth task from meta-testing set; $D_{s,j}^{test}$: the support set of T_j^{test}; $D_{q,j}^{test}$: the query set of T_j^{test}; α_{inner}: learning rate of the inner loop; α_{outer}: learning rate of the outer loop; θ: shared meta-parameters; φ: task embeddings

initialize θ and φ
while not done **do**
    Sample batch of meta-tasks $\{T_i^{train}\}_{i=1}^{M}$ from D^{train}
    for each T_i^{train} in $\{T_i^{train}\}_{i=1}^{M}$ **do**
        Evaluate $\nabla_\varphi \mathcal{L}(\theta, \varphi, D_{s,i}^{train})$ by MSE loss
        update $\varphi_i \leftarrow \varphi - \alpha_{inner} \cdot \nabla_\varphi \mathcal{L}(\theta, \varphi, D_{s,i}^{train})$
    end for
    Evaluate $\sum_{i=1}^{M} \mathcal{L}(\theta, \varphi_i, D_{q,i}^{train})$ by MSE loss
    update $\theta \leftarrow \theta - \alpha_{outer} \cdot \nabla_\theta \frac{1}{M} \sum_{i=1}^{M} \mathcal{L}(\theta, \varphi_i, D_{q,i}^{train})$
end while
for each T_j^{test} in D^{test} **do**
    Evaluate $\nabla_\varphi \mathcal{L}(\theta, \varphi, D_{s,j}^{test})$ by MSE loss
    update $\varphi_j \leftarrow \varphi - \alpha_{inner} \cdot \nabla_\varphi \mathcal{L}(\theta, \varphi, D_{s,j}^{test})$
    Perform RUL estimation for $D_{q,j}^{test}$
end for

4. Experimental case study 1

4.1. Data description

In this case study, various experiments were performed on a benchmarking problem regarding turbofan engine RUL estimation, i.e. C-MAPSS datasets (Saxena et al., 2008). The datasets are provided by the NASA Ames Prognostics Data Repository through Commercial Modular Aero-Propulsion System Simulation, which involves 4 subsets: FD001, FD002, FD003, and FD004. As shown in Table 1, each subset is composed of multiple multi-variate temporal trajectories that are divided into training trajectories and testing trajectories, and the multi-variate trajectory is comprised of 21 sensor values and 3 setting parameters. Training trajectories contain the full lifecycle monitoring data of engines and testing trajectories are truncated before failure.

To validate our framework in the cross-domain few-shot situation where the trajectories of the base domain are sufficient and the trajectories of the novel domain are limited, typical 12 cross-domain scenarios that focus on a single base domain were set up for the experiments, as shown in

Table 2. In FD001 and FD002, the failure comes from the high-pressure compressor (HPC). In FD003 and FD004, the failure comes from the HPC and fan. In FD002 and FD004, degradation occurs under variable operational conditions. For the base domain, a series of run-to-failure trajectories were randomly selected from the corresponding training set to construct the meta-tasks. For the novel domain, K run-to-failure trajectories were randomly selected from the corresponding training set to form the support set, and all truncation points in the corresponding test set were used to form the query set for evaluation.

For the C-MAPSS datasets, a piece-wise linear degradation model has been proven to be feasible and effective. The degradation model follows the assumption that the RUL label RUL_{early} remains constant (125 cycles) at the initial stage before decreasing linearly with engine degradation until failure.

To evaluate the effectiveness of the proposed framework in RUL estimation, the root mean square error (RMSE) equation was utilized as a performance metric:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d_i^2}, \quad d_i = R\widehat{U}L_i - RUL_i, \tag{6}$$

where RUL_i and $R\widehat{U}L_i$ represent the actual RUL and estimated RUL for the ith query sample in the meta-testing set, respectively. Additionally, N denotes the total amount of query samples in the meta-testing set. The other metric that was used is the Score function, which imposes more substantial penalties for late predictions:

$$Score = \sum_{i=1}^{N} s_i, \quad s_i = \begin{cases} e^{-d_i/13} - 1, d_i < 0 \\ e^{d_i/10} - 1, d_i \geq 0 \end{cases}. \tag{7}$$

Similar to the RMSE, RUL estimation frameworks with better performance will produce lower Score value. However, unlike the RMSE, the Score function assigns heavier penalties for late predictions ($d_i > 0$) compared to early predictions ($d_i < 0$) because late predictions are more likely to cause serious catastrophes and massive economic losses.

4.2. Data preprocessing

4.2.1. Data normalization

The numerical ranges of various sensors are significantly different, which severely hinders convergence of the predictive model. In this situation, normal Min-Max normalization is typically employed to normalize the data within the range of [-1, 1]. However, for variable operational conditions, the numerical ranges of the monitoring data from the same sensor but different operational conditions are still different, which hinders the extraction of the degradation tendency. Consequently, the sensor values were normalized within the range of [-1, 1] by performing variable operational conditions using Min-Max normalization:

$$\tilde{x}_i^{j,k} = \frac{2(x_i^{j,k} - x_{\min}^{j,k})}{x_{\max}^{j,k} - x_{\min}^{j,k}} - 1. \tag{8}$$

Table 2
Typical 12 cross-domain scenarios.

	Base		Novel		Base		Novel
Scenario (1)	FD001	→	FD003	Scenario (7)	FD003	→	FD004
Scenario (2)	FD003	→	FD001	Scenario (8)	FD004	→	FD003
Scenario (3)	FD002	→	FD004	Scenario (9)	FD002	→	FD003
Scenario (4)	FD004	→	FD002	Scenario (10)	FD003	→	FD002
Scenario (5)	FD001	→	FD002	Scenario (11)	FD001	→	FD004
Scenario (6)	FD002	→	FD001	Scenario (12)	FD004	→	FD001

Table 3
Hyperparameters of the model structure.

Parameter	Values
Input size	30×14
Conv 1	kernel: 3×1 channel: n_C
Conv 2	kernel: 3×1 channel: n_C
Conv 3	kernel: 3×1 channel: n_C
Conv 4	kernel: 3×3 channel: n_C
Linear 1	$336n_C \times 1$
Embeddings	n_E
Linear 2	$n_E \times 2n_E$
Linear 3	$2n_E \times 2n_E$
Output size	1

Here, k indicates the operational condition category, $x_i^{j,k}$ represents the original sensor value of the ith time point and jth sensor, and $\tilde{x}_i^{j,k}$ is the normalized sensor value of $x_i^{j,k}$. Additionally, $x_{\max}^{j,k}$ and $x_{\min}^{j,k}$ denote the maximum and minimum original values from the jth sensor under the operational condition k, respectively. The k-means algorithm was then leveraged according to 3 setting parameters to classify the sensor data of each time point. After normalization, the values from some of the sensors remained constant or fluctuated around a constant value throughout the full lifecycle without any degradation information. Therefore, 14 sensors with a significant degradation trend were selected, similar to a previous study (Li et al., 2018).

4.2.2. Sliding time window

Multi-time step data contain considerably more time series information than single time step data; thus, the sliding time window was exploited to obtain multi-time step data. Given the time window size W, multi-variate data from the $i - W$th step to the ith step were enclosed as the feature map of the ith step, where the shape of each feature map is $[W, 14]$. To balance the estimation performance and computing load, the time window size W was set to 30, and the trajectories whose maximum record cycles are less than 30 were disregarded.

4.3. Experimental settings

In this case, K was set to 1 and 30 % of the training trajectories were randomly selected from the base domain to construct the meta-tasks, which were enriched via segmentation with P = 30.

As shown in Fig. 4 and Fig. 5, the preprocessed data were put into the feature extractor, and the transformation of the embeddings was connected to the third convolutional layer of the feature extractor via FiLM. After the feature extractor, the feature maps were flattened and the RUL values were output through a fully-connected layer of the regressor. Table 3 lists the detailed model structure hyperparameters. When the base domain was either FD001 or FD003, the values of both the n_C and n_E were set to 50. When the base domain was either FD002 or FD004, the values of both the n_C and n_E were set to 100. Notably, zeros-padding operation was implemented to keep the feature map shape unchanged for the first three convolutional layers.

For the meta-learning process, the embeddings were trained using a 1-gradient inner step for both the meta-training and meta-testing phases. Additionally, the initial parameters of the embeddings φ were set to $[0, \ldots, 0]^\top$, the outer learning rate was set to 5e-5, and the inner learning rate was set to 2e-5. The Adam optimizer was utilized to perform optimization, and the reported experimental results were averaged over 50 repetitions to attenuate the randomness of sampling and training.

4.4. Ablation study

Compared with the cross-domain few-shot RUL estimation frameworks that employ the traditional MAML, the segmentation strategy and task embeddings are introduced into our framework. Therefore, ablation

Table 4
Ablation studies of the segmentation strategy and task embeddings in FD001→FD003, FD003→FD001, FD002→FD004, and FD004→FD002.

Ablation cases	Segmentation	Task embeddings	FD001→FD003		FD003→FD001		FD002→FD004		FD004→FD002	
			RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
(1)			26.22	3196	27.67	2534	29.31	9053	25.91	6159
(2)	✓		26.07	2621	25.10	1973	28.96	8824	24.53	3675
(3)		✓	24.59	1902	27.34	2380	28.29	7947	25.56	5072
(4)	✓	✓	**24.34**	**1829**	**24.24**	**1605**	**26.96**	**6944**	**23.24**	**2651**

Table 5
Ablation studies of the segmentation strategy and task embeddings in FD001→FD002, FD002→FD001, FD003→FD004, and FD004→FD003.

Ablation cases	Segmentation	Task embeddings	FD001→FD002		FD002→FD001		FD003→FD004		FD004→FD003	
			RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
(1)			35.74	12,072	33.45	5669	36.12	12,349	34.83	5782
(2)	✓		34.61	9413	32.11	4695	35.02	9034	33.74	4711
(3)		✓	34.47	9146	32.78	4792	34.88	8879	34.13	5017
(4)	✓	✓	**32.06**	**7678**	**30.89**	**4036**	**32.37**	**7855**	**31.70**	**4145**

Table 6
Ablation studies of the segmentation strategy and task embeddings in FD002→FD003, FD003→FD002, FD001→FD004, and FD004→FD001.

Ablation cases	Segmentation	Task embeddings	FD002→FD003		FD003→FD002		FD001→FD004		FD004→FD001	
			RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
(1)			37.09	6453	37.26	13,524	37.68	13,771	36.70	6194
(2)	✓		35.71	5538	35.45	9595	36.26	11,076	34.42	5136
(3)		✓	35.52	5175	36.51	10,764	35.82	10,631	35.25	5578
(4)	✓	✓	**33.21**	**4534**	**33.16**	**8965**	**33.43**	**9366**	**32.71**	**4377**

experiments were performed to verify their validity and necessity. Consequently, four ablation cases are discussed: (1) neither the segmentation strategy nor task embeddings are used, that is, MAML algorithm without the segmentation strategy. (2) Only the segmentation strategy is used, that is, MAML algorithm with the segmentation strategy. (3) Only task embeddings are used, that is, MAML with the task embeddings algorithm and without the segmentation strategy. (4) Both the segmentation strategy and task embeddings are used, that is, MAML with the task embeddings algorithm and segmentation strategy. As shown in Table 4, Table 5, and Table 6, the addition of each part can effectively improve the estimation performance, and the predictive model produces the best estimation performance when both parts are employed together, which indicates that the two parts are indispensable.

For the segmentation strategy, the number of meta-tasks can be increased by 3–5 times, which enables the model to capture more generalized meta degradation knowledge, thereby effectively improving the estimation performance. For the task embeddings, their utilization leads to a reduction in the number of required parameter updates of about 30–70 % during the adaptation process compared with the framework without task embeddings. As mentioned above, task embeddings effectively mitigate overfitting by performing updates with extremely few labeled data on a relatively low-dimensional parameter space, significantly improving the estimation performance. Meanwhile, the enhanced performance also benefits from the adoption of representation change because it significantly improves cross-domain adaptability.

Table 7
Details of JT9D Engine Datasets.

Sub-dataset	SDM	MDM
Time series training set	150	150
Time series test set	50	50

5. Experimental case study 2

5.1. Data description

In the second case study, various experiments were conducted on new aero-turbofan engine degradation datasets based on the JT9D engine. A set of thermodynamic mechanism simulation models of the aero-turbofan engine are established to build the datasets. The standard parameters of the engine and its components are obtained from the Toolbox for the Modeling and Analysis of Thermodynamic Systems (Chapman et al., 2014) developed by NASA. As depicted in Table 7, the datasets involve 2 subsets with different fault modes, denoted as the single degradation mode (SDM) and multiple degradation mode (MDM). Each multi-variate trajectory is comprised of 26 sensor values.

To validate our framework in the cross-domain few-shot situation where the trajectories of the base domain are sufficient and the trajectories of the novel domain are limited, the cross-domain scenario SDM→MDM was set up for the experiments. In the SDM, the failures come from the efficiency degradation of the HPC. In the MDM, the failures come from the efficiency and flow degradation of the HPC. For the base domain, a series of run-to-failure trajectories were randomly selected from the corresponding training set to construct the meta-tasks. For the novel domain, K run-to-failure trajectories were randomly selected from the corresponding training set to form the support set, and

Table 8
Ablation studies of the segmentation strategy and task embeddings in SDM→MDM.

Ablation cases	Segmentation	Task embeddings	SDM→MDM	
			RMSE	Score
(1)			9.66	69.74
(2)	✓		9.01	61.97
(3)		✓	8.58	55.31
(4)	✓	✓	**7.72**	**41.05**

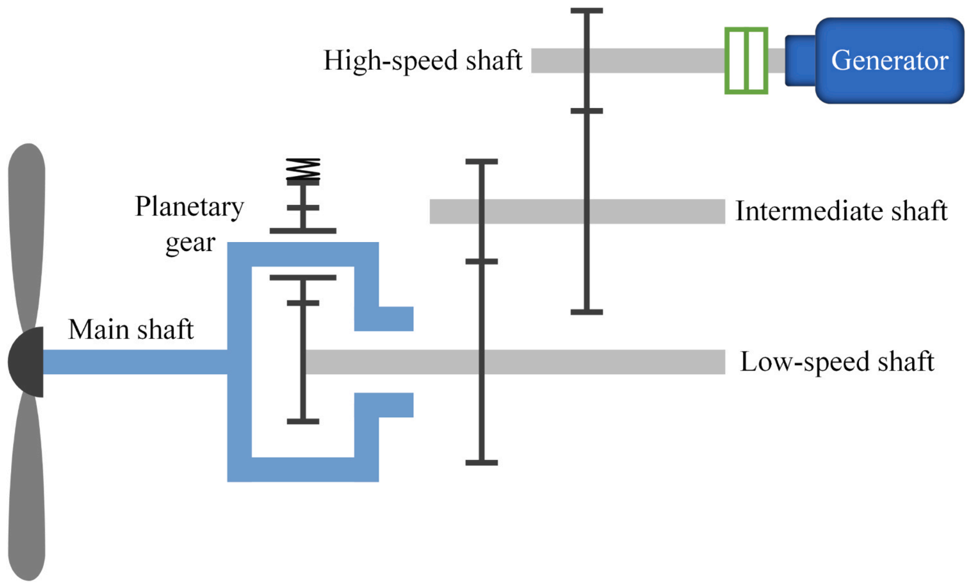

Fig. 6. The simplified structure of the wind turbine gearbox.

all truncation points in the test set were used to form the query set for evaluation. Meanwhile, to evaluate the effectiveness of the proposed framework, the RMSE and Score were utilized as the performance metrics.

5.2. Data preprocessing and experimental settings

For data preprocessing, the data normalization and sliding time window were also employed, similar to case study 1. For the experimental settings, the embeddings were trained using the 1-gradient inner step for the meta-training and meta-testing phases. Additionally, the outer and inner learning rate were set to 1e-3, and the other settings were kept consistent with those of the FD001→FD003 scenario in case study 1.

5.3. Ablation study

To verify the validity and necessity of the segmentation strategy and task embeddings, ablation experiments were performed as shown in Table 8, where the ablation cases are identical to those discussed in Section 4.4. It can be observed that the utilization of each part effectively improves the estimation performance, which further demonstrates the indispensability of both components.

6. Experimental case study 3

6.1. Data description

To demonstrate the practical applicability of the proposed framework within real-world prognostics and health management workflows, its application in an industrial wind turbine gearbox condition monitoring system was considered. The transmission system of industrial wind turbine gearboxes primarily consists of the main shaft, a first-stage planetary gear train, and a second-stage parallel gear train. The simplified structure of the wind turbine gearbox is illustrated in Fig. 6, where the gearbox type is FL2000H and the total transmission ratio is 117.66. To ensure long-term stability and facilitate real-time fault diagnosis and prognosis, sensors are deployed on the gearboxes, enabling the continuous collection of full lifecycle operational data. The integration of the proposed framework into the system follows a three-stage process:

(i) **Initial model deployment:** Historical degradation trajectories collected by the monitoring system are used to train a foundational predictive model with meta-knowledge, which is then deployed within the condition monitoring system.

(ii) **Model adaptation:** During operation, the wind turbine may encounter operating profiles that were not previously observed in historical degradation trajectories. As failures occur, new degradation trajectories are collected, allowing the foundational predictive model to be refined based on deterioration patterns that actually occur in current new profiles.

(iii) **Real-time prognostic and maintenance decision support:** The refined predictive model is deployed within the real-time monitoring system, utilizing sensor measurements to estimate the RUL of the wind turbine. The prediction results are then leveraged to implement proactive maintenance strategies, reducing unexpected failures and enhancing the reliability of wind turbine operations.

In specific experiments, these sensors are deployed on the gearboxes to record data from 9 sensors and 3 setting parameters. To validate our framework in a cross-domain few-shot scenario, the cross-domain scenario BD→ND was established for experimentation. "BD" (base domain) consists of multiple historical degradation trajectories, and "ND" (novel domain) contains newly-observed degradation trajectories. For the base domain, a series of historical trajectories were randomly selected to construct meta-tasks. In the novel domain, K run-to-failure trajectories were utilized to form the support set, and real-time operating equipment monitoring data were used as the query set for evaluation. Meanwhile, the RMSE and Score were utilized as the performance metrics to evaluate the effectiveness of the proposed framework.

Table 9
Ablation studies of the segmentation strategy and task embeddings in BD→ND.

Ablation cases	Segmentation	Task embeddings	BD→ND	
			RMSE	Score
(1)			0.2217	3.58
(2)	✓		0.1998	3.19
(3)		✓	0.2063	3.35
(4)	✓	✓	**0.1832**	**3.01**

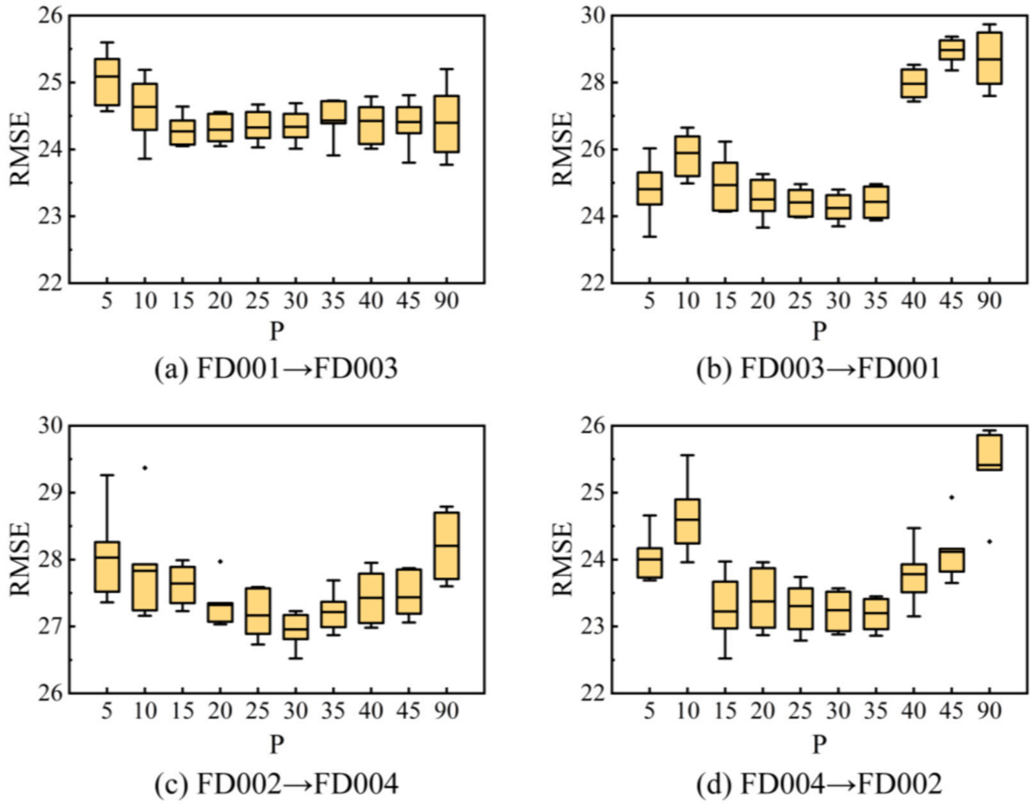

Fig. 7. The RMSE values of the proposed framework with different P in the case study 1.

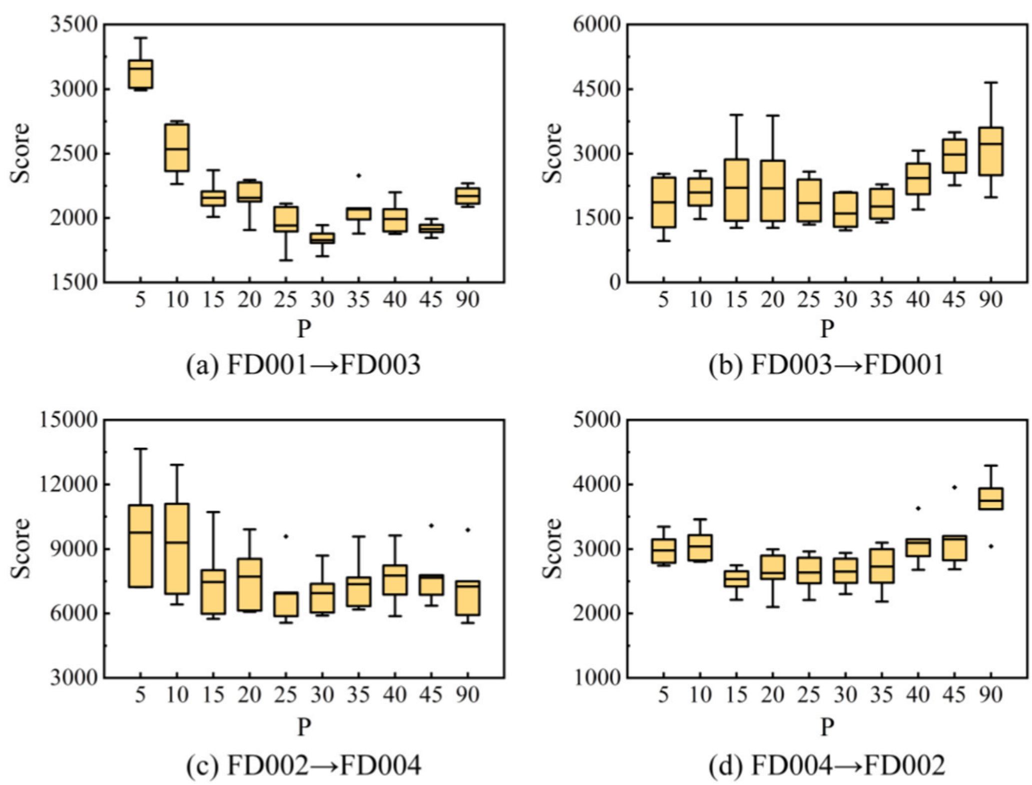

Fig. 8. The Score values of the proposed framework with different P in the case study 1.

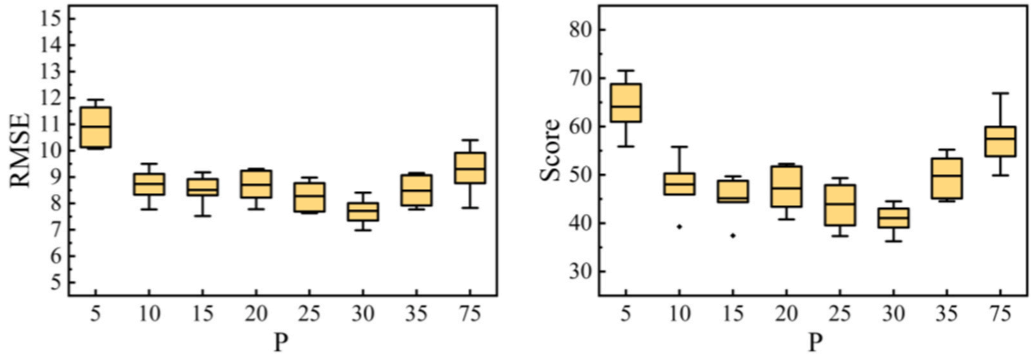

Fig. 9. The RMSE and Score values of the proposed framework with different P in the case study 2.

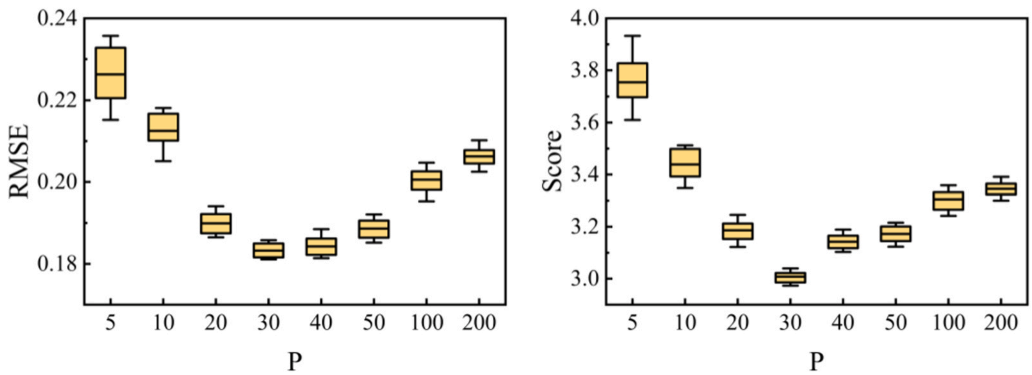

Fig. 10. The RMSE and Score values of the proposed framework with different P in the case study 3.

6.2. Data preprocessing and experimental settings

For data preprocessing and the experimental settings, the outer and inner learning rate were set to 1e-4 and the other settings were kept consistent with those of the FD001→FD003 scenario in case study 1. Additionally, label normalization was applied to normalize the RUL values within the range of [0, 1], thereby minimizing differences between trajectories.

6.3. Ablation study

To validate the effectiveness and necessity of the segmentation strategy and task embeddings, ablation experiments were conducted, as presented in Table 9. The ablation cases analyzed here are similar to those discussed in Section 4.4. The results indicate that incorporating each component significantly enhances estimation performance, further highlighting the essential role of both elements.

7. Experimental analysis

7.1. Analysis of the segmentation strategy

As depicted in Section 3.2.1, the segmentation strategy can effectively increase the number of meta-tasks, enabling the model to capture more generalized meta degradation knowledge. However, the selection of the number of segments P may have an important effect on the estimation performance. To analyze the effect and efficiently use the data provided, different P were applied to our frameworks in the case studies. Drawing on the four prior scenarios as illustrative cases, box plots of the RMSE and Score values with different P were obtained, as shown in Fig. 7, Fig. 8, Fig. 9, and Fig. 10. The growth of the P, RMSE and Score values tend to descend first before ascending. This might be because one trajectory can only generate one or two sub-trajectories when P is too large, resulting in few meta-tasks and thus limiting the meta-training process. When P is too small, one trajectory can generate many sub-trajectories; however, each sub-trajectory contains few time points,

Table 10
Comparison experiments of different algorithms in FD001→FD003, FD003→FD001, FD002→FD004, and FD004→FD002.

Algorithms	FD001→FD003		FD003→FD001		FD002→FD004		FD004→FD002	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
Pretrain-finetune w/o freezing	33.78	12,709	27.50	3173	32.61	30,854	25.54	12,691
Pretrain-finetune w/ freezing	33.37	8610	25.75	2264	33.99	26,829	25.72	11,619
ANIL	26.04	2593	25.36	2046	28.89	8517	24.96	3979
AMAML	26.22	2771	25.55	2336	29.02	7812	25.17	4373
MTL	26.74	2836	25.32	2377	29.24	9755	24.84	4785
ALFA	26.23	2751	25.93	2289	29.84	8219	25.32	3956
Gradient-Agreement	27.98	2690	26.47	2184	29.33	7893	25.33	5257
Reptile	26.21	3351	25.55	2486	31.87	18,047	26.93	15,515
MAML	26.07	2621	25.10	1973	28.96	8824	24.53	3675
MAML w/ task embeddings	**24.34**	**1829**	**24.24**	**1605**	**26.96**	**6944**	**23.24**	**2651**

which leads to limited information on each sub-trajectory. It can also be observed that the estimation performance is generally better when P is between 25 and 35.

7.2. Analysis of the task embeddings

As depicted in Section 3.2.2, after employing task embeddings, the proposed training algorithm can effectively mitigate overfitting and adopt representation change as the main adaptation strategy to improve cross-domain adaptability. Thus, a series of experimental analyses is outlined in this section to clearly illustrate the benefits of task embeddings.

7.2.1. Prognostic performance

To illustrate the effectiveness and superiority of the proposed training algorithm using task embeddings, a series of experiments was conducted to compare our algorithm with several existing cross-domain few-shot RUL estimation training algorithms. Notably, the experiments focus on the comparison of the cross-domain few-shot training algorithms rather than model structures. Therefore, the cross-domain few-shot RUL estimation training algorithms adopted in the different literatures were compared under the premise of using the segmentation strategy and the same convolutional neural network as the model structure. Meanwhile, for the fairness of comparison, the scope of the algorithms was limited to inductive TL to ensure that all the comparison algorithms can make use of the supervised degradation information from the novel domain. Specifically, pretrain-finetune used in Zhang et al. (2018), the original MAML used in Ding et al. (2021); Mo et al. (2023) and the proposed algorithm were employed for comparison. Moreover, the commonly used meta-learning training algorithms such as Reptile (Nichol et al., 2018), Gradient-Agreement (Eshratifar et al., 2018), ALFA (Baik et al., 2020a), MTL (Eshratifar et al., 2019), AMAML (Pan et al., 2022), and ANIL (Shao et al., 2024) were also introduced for comparison.

The experimental results of the above algorithms in some scenarios of case study 1 are shown in Table 10. The overall performance of the proposed algorithm is better than that obtained in others. In comparison, the mean RMSE and mean Score of our algorithm decrease significantly, which verifies the effectiveness of our algorithm. Meanwhile, the meta-learning-based algorithms achieve promising performance in comparison with the pretrain-finetune-based algorithms. More specifically, the descending percentages of the mean RMSE and mean Score are 11 % and 63 %, respectively. This is because the unique mechanism of meta-learning allows the algorithms to have a greater generalization capability on new tasks in the case of limited labeled samples. Similarly, as shown in Table 11, Table 12, Table 13, and Table 14, the comparison results in other scenarios can also serve as supplementary evidence to support the aforementioned analysis.

7.2.2. Impact of the connection position

To analyze the impact of the position where task embeddings are connected to the backbone network, several experiments with different connection positions were implemented, with FD001→FD003, FD003→FD001, FD002→FD004, and FD004→FD002 as examples. Fig. 11 shows that as the connection position moves from the first to the fourth convolutional layer, the estimation performance first improves before declining. The main reason is that the early convolution layers produce the task-invariant features that are shared across both the base and novel domains, while the later convolution layers produce discriminative task-specific features for better task adaptation (Zeiler and Fergus, 2014). According to the experimental results, the third convolution layer is the optimal position for task adaptation when responding to different domains. In addition, the estimation performance with the connection position in the last three convolution layers is better than that with the connection position in the fully-connected layer. This suggests that representation change at a suitable position can outperform representation reuse in cross-domain RUL estimation scenarios.

7.2.3. Expanded statistical analysis

To provide a statistical foundation for our findings, detailed evaluations of both the standard deviation (STD) and coefficient of variation (CV) were conducted. The standard deviation is a measure that describes the dispersion of the data points from the mean:

$$STD = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Metric_i - \mu)^2}, \mu = \frac{1}{n}\sum_{i=1}^{n}Metric_i, \tag{9}$$

where $Metric_i$ represents the performance metric (either RMSE or Score) for the ith repetition, and n denotes the total amount of repetitions. Additionally, CV is a relative measure that standardizes the standard deviation using the mean, which is particularly useful for describing the degree of the dispersion at different scales:

$$CV = \frac{STD}{\mu} \times 100\%. \tag{10}$$

As shown in Table 15 and Table 16, statistical experiments were implemented, where FD001→FD003, FD003→FD001, FD002→FD004, and FD004→FD002 were used as examples. It is noteworthy that the CV values for both the RMSE and Score metrics within the proposed framework consistently remain below 20 % in most cross-domain scenarios, which underscores the robustness of the framework and affirms the reliability of the experimental results. Additionally, compared to other algorithms, the proposed algorithm exhibits lower STD and CV values across the majority of the cross-domain scenarios, effectively demonstrating its advantages.

To visually analyze the robustness, a box plot of the RMSE is presented in Fig. 12. Meta-learning-based and pretrain-finetune-based algorithms exhibit similar robustness in FD003→FD001 and FD004→FD002. Nevertheless, in FD001→FD003 and FD002→FD004, pretrain-finetune-based algorithms are considerably less robust than meta-learning-based algorithms. This discrepancy is primarily owing to

Table 11
Comparison experiments of different algorithms in FD001→FD002, FD002→FD001, FD003→FD004, and FD004→FD003.

Algorithms	FD001→FD002		FD002→FD001		FD003→FD004		FD004→FD003	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
Pretrain-finetune w/o freezing	42.71	24,151	41.6	11,620	43.73	28,487	42.59	13,746
Pretrain-finetune w/ freezing	41.8	21,753	41.14	9722	42.28	23,856	41.7	11,484
ANIL	34.73	9551	31.97	4520	35.16	9986	33.41	4683
AMAML	35.44	12,099	33.02	5074	35.94	13,950	34.53	4990
MTL	35.76	12,847	33.58	5322	36.62	13,623	34.91	5226
ALFA	35.12	11,316	33.26	5239	35.89	13,145	34.65	5124
Gradient-Agreement	36.27	10,868	34.79	4971	37.03	12,232	35.73	5018
Reptile	34.83	12,552	33.24	5877	35.59	20,756	34.62	5991
MAML	34.61	9413	32.11	4695	35.02	9034	33.74	4711
MAML w/ task embeddings	**32.06**	**7678**	**30.89**	**4036**	**32.37**	**7855**	**31.70**	**4145**

Table 12
Comparison experiments of different algorithms in FD002→FD003, FD003→FD002, FD001→FD004, and FD004→FD001.

Algorithms	FD002→FD003		FD003→FD002		FD001→FD004		FD004→FD001	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
Pretrain-finetune w/o freezing	44.92	15,855	44.87	36,059	45.39	39,376	44.35	14,233
Pretrain-finetune w/ freezing	42.64	12,782	43.93	28,991	43.47	32,519	43.79	12,232
ANIL	35.93	5889	35.32	9456	36.41	11,378	34.29	4991
AMAML	36.85	6425	36.23	12,662	37.25	13,381	35.31	5313
MTL	36.77	6576	36.64	14,187	37.38	15,610	35.97	6160
ALFA	36.5	5935	36.27	13,372	37.1	14,473	35.65	5668
Gradient-Agreement	36.99	5183	37.15	11,826	37.65	12,755	36.34	5132
Reptile	36.12	6741	36.76	16,534	36.74	23,443	35.14	6369
MAML	35.71	5538	35.45	9595	36.26	11,076	34.42	5136
MAML w/ task embeddings	**33.21**	**4534**	**33.16**	**8965**	**33.43**	**9366**	**32.71**	**4377**

Table 13
Comparison experiments of different algorithms in SDM→MDM.

Algorithm	SDM→MDM	
	RMSE	Score
Pretrain-finetune w/o freezing	12.26	93.93
Pretrain-finetune w/ freezing	12.55	96.59
ANIL	9.86	65.83
AMAML	10.53	66.2
MTL	11.07	70.31
ALFA	10.62	66.08
Gradient-Agreement	11.26	73.74
Reptile	11.87	74.62
MAML	9.01	61.97
MAML w/ task embeddings	**7.72**	**41.05**

Table 14
Comparison experiments of different algorithms in BD→ND.

Algorithm	BD→ND	
	RMSE	Score
Pretrain-finetune w/o freezing	0.2466	3.93
Pretrain-finetune w/ freezing	0.2391	3.69
ANIL	0.2083	3.43
AMAML	0.2012	3.26
MTL	0.2195	3.51
ALFA	0.2120	3.39
Gradient-Agreement	0.2174	3.52
Reptile	0.2287	3.61
MAML	0.1998	3.19
MAML w/ task embeddings	**0.1832**	**3.01**

the presence of only one fault mode in FD001 and FD002, whereas FD003 and FD004 feature two fault modes. Therefore, in FD003→FD001 and FD004→FD002, the HPC degradation in the novel domain has appeared in the base domain, significantly reducing the difficulty of cross-domain adaptation. However, in FD001→FD003 and FD002→FD004, the fan degradation in the novel domain does not exist in the base domain, resulting in larger domain divergences. This notable robustness advantage in scenarios with slightly higher cross-domain difficulty, such as FD001→FD003 and FD002→FD004, highlights the reliability of meta-learning algorithms in challenging scenarios.

To further evaluate the RUL estimation performance of the proposed algorithm from a statistical analysis perspective, the error distribution of the RUL estimation values on the test set were computed, as illustrated in Fig. 13. As shown in the figure, the error follows a distribution close to normal. The majority of the errors are concentrated within the smaller error interval, with only a few instances falling within the higher error interval. Meanwhile, the distribution center exhibits a slight shift towards negative values. This suggests that the model tends to perform early predictions, which contributes to the prevention of severe incidents and the reduction of potential economic losses.

To further investigate the behavioral patterns of the time series data, we analyzed the temporal trends in the absolute error and confidence interval (CI) width, as shown in Fig. 14. As time progresses and the engine approaches failure, both the absolute error and CI width of the proposed algorithm decrease significantly. This suggests that the algorithm exhibits an improved estimation accuracy and robustness in this phase, thereby supporting the need for timely maintenance decisions as failure approaches.

7.2.4. Discussion on the data size

To demonstrate the merits of the proposed algorithms when using very few samples, the impact of the data size on various algorithms was investigated, taking FD001→FD003 as an example. To analyze the impact of the novel domain data size, several experiments were performed with K values of 5, 2, and 1, and the RMSE results are displayed in Fig. 15 (a). As K decreases, the RMSE of the proposed algorithm slightly increases as the RMSE of the other algorithms drastically increases. This is mainly because the proposed algorithm only updates task embeddings φ in the inner loop and adaptation process, attenuating overfitting in scenarios with limited data. The results demonstrate that the proposed algorithm has a greater advantage over the others when there is less data in the novel domain.

To analyze the impact of the base domain data size, several experiments were performed with 10 %, 30 %, and 60 % base domain trajectories, and the RMSE results are shown as Fig. 15 (b). It can be observed that the RMSE values increase with the reduction in the base domain data size owing to the fact that the information extracted in the pretraining or meta-training phase suffers from a reduced data quantity of the base domain, which indicates that the base domain data quantity positively influences the estimation quality.

7.3. Comparison experiments

In this section, we demonstrate the effectiveness and superiority of the proposed framework by comparing it with several established prognostic frameworks from previous literatures, including DSSANN (Xu et al., 2022), DCRN (Shang et al., 2024), LSTM-DANN (Costa et al., 2020), LSTM-ADDA (Duan et al., 2022), LSTM-FT (Zhang et al., 2018), MAML (Mo et al., 2023). Table 17, Table 18, and Table 19 present the comparison results on the public C-MAPSS datasets, where “imp” highlights the improvements of the proposed framework over existing advanced prognostic frameworks.

It is found that DSSANN and DCRN exhibit a relatively poor estimation performance as they do not account for the distribution shifts that invariably exist across different domains. LSTM-DANN and LSTM-ADDA employ unsupervised domain adaptation techniques, which enhances the cross-domain capability of the models. However, their predictive capabilities are somewhat limited compared to LSTM-FT and MAML owing to their inability to utilize the small amounts of labeled data available in novel domains. In comparison to these established prognostic frameworks, the proposed framework not only leverages the

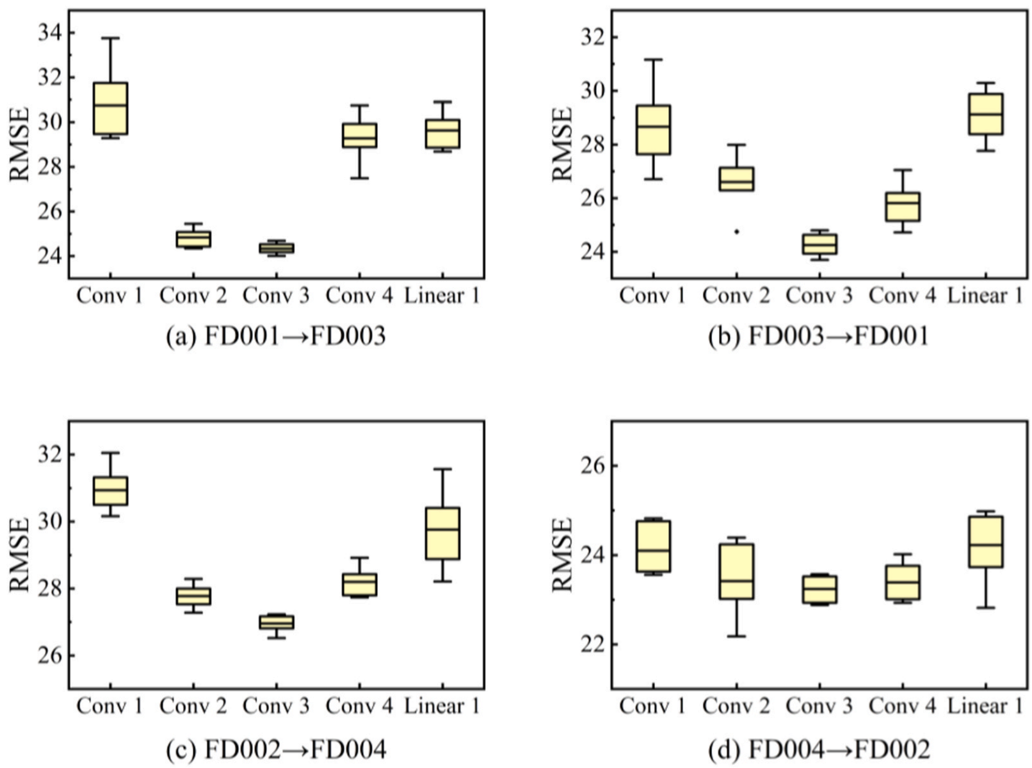

Fig. 11. Experimental results with different connection position.

Table 15
Statistical experiments of the RMSE.

Algorithm	FD001→FD003		FD003→FD001		FD002→FD004		FD004→FD002	
	STD	CV	STD	CV	STD	CV	STD	CV
Pretrain-finetune w/o freezing	2.88	8.54 %	1.27	4.63 %	2.72	8.33 %	1.19	4.64 %
Pretrain-finetune w/ freezing	2.72	8.15 %	0.93	3.61 %	2.40	7.05 %	0.75	2.92 %
ANIL	0.48	1.82 %	0.70	2.78 %	0.46	1.58 %	0.67	2.70 %
AMAML	0.41	1.58 %	0.88	3.43 %	0.68	2.34 %	0.79	3.13 %
MTL	0.46	1.72 %	0.86	3.31 %	0.74	2.50 %	0.72	2.85 %
ALFA	0.69	2.65 %	0.93	3.67 %	0.83	2.84 %	0.83	3.36 %
Gradient-Agreement	0.84	3.00 %	0.87	3.28 %	0.84	2.85 %	0.76	3.01 %
Reptile	0.96	3.67 %	1.12	4.39 %	0.63	1.99 %	0.80	2.97 %
MAML	0.39	1.50 %	0.66	2.62 %	0.46	1.59 %	0.66	2.69 %
MAML w/ task embeddings	**0.27**	**1.12 %**	**0.46**	**1.91 %**	**0.29**	**1.08 %**	**0.32**	**1.39 %**

Table 16
Statistical experiments of the Score.

Algorithm	FD001→FD003		FD003→FD001		FD002→FD004		FD004→FD002	
	STD	CV	STD	CV	STD	CV	STD	CV
Pretrain-finetune w/o freezing	9732	76.58 %	1011	31.88 %	18,682	60.55 %	2634	20.76 %
Pretrain-finetune w/ freezing	3230	37.51 %	551	24.34 %	11,306	42.14 %	3577	30.79 %
ANIL	458	17.65 %	466	22.79 %	936	10.98 %	771	19.38 %
AMAML	655	23.64 %	556	23.79 %	1028	13.15 %	1694	38.73 %
MTL	407	14.35 %	301	12.66 %	1012	10.37 %	755	15.77 %
ALFA	605	22.00 %	467	20.41 %	1214	14.77 %	1534	38.77 %
Gradient-Agreement	389	14.46 %	637	29.16 %	1170	14.82 %	1700	32.33 %
Reptile	951	28.39 %	**242**	**9.76 %**	2117	11.73 %	2173	14.01 %
MAML	419	15.97 %	382	19.36 %	**803**	**9.10 %**	640	17.41 %
MAML w/ task embeddings	**183**	**9.99 %**	452	28.19 %	1139	16.41 %	**347**	**13.07 %**

unique mechanism of meta-learning, which facilitates generalization in cross-domain few-shot scenarios, but also further enhances its estimation performance through the introduction of a segmentation strategy, task embedding design, and representation change. These enhancements effectively prevent overfitting and improve cross-domain adaptability, allowing the proposed framework to consistently exhibit optimal performance across various cross-domain scenarios.

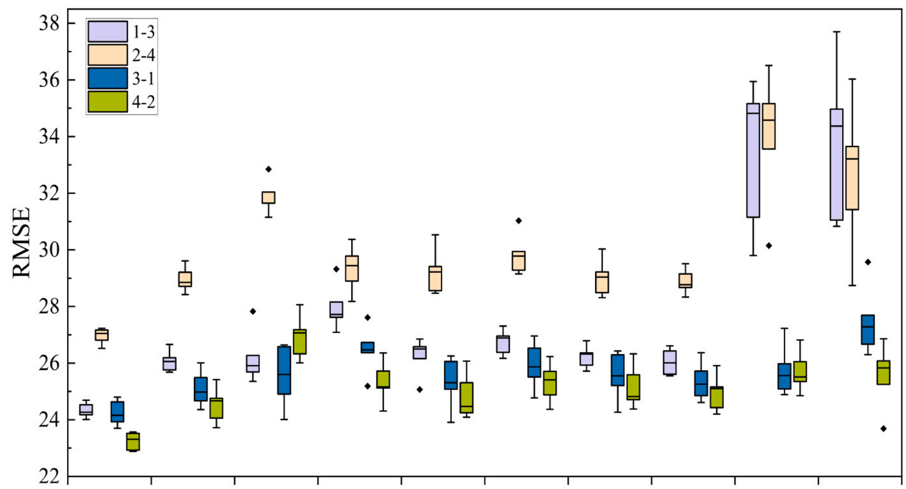

Fig. 12. Box plot of the RMSE.

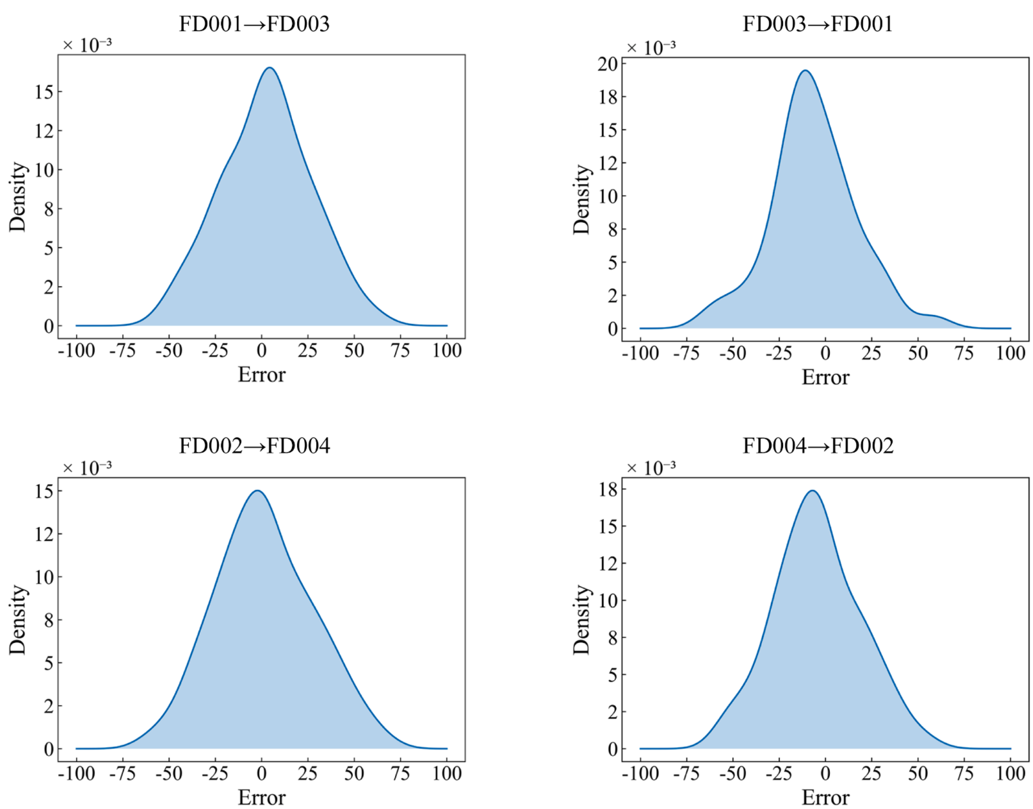

Fig. 13. Estimation error distribution.

8. Discussion

The proposed cross-domain few-shot prognostic framework demonstrates evident application potential across various industrial domains. In aviation applications, the cost of aircraft engines and critical structural components is extremely high, resulting in limited labeled data. The ability to refine predictive models with minimal labeled data is particularly valuable for real-time health monitoring and failure prediction in aviation maintenance systems. In the wind energy sector, the environmental factors are highly diverse, leading to different failure modes and impact the overall performance and reliability of the turbines. Adapting predictive models to new operating profiles with limited labeled data is essential for effectively predicting faults and scheduling maintenance in wind turbine systems. The potential of our framework in these domains has been preliminarily validated through the three case studies discussed above. Beyond these applications, the proposed cross-domain few-shot prognostic framework has broader applicability across other industrial domains. In manufacturing, complex production machinery such as computer numerical control (CNC) machines and robotic arms often encounter wear-related faults that evolve over time. The proposed framework can be integrated into smart factory environments where predictive maintenance strategies rely on sensor data collected across different operating profiles. Furthermore, in transportation systems, such as railways and autonomous vehicles, drivetrain components and braking systems undergo diverse degradation patterns under different operating profiles. The ability to transfer knowledge

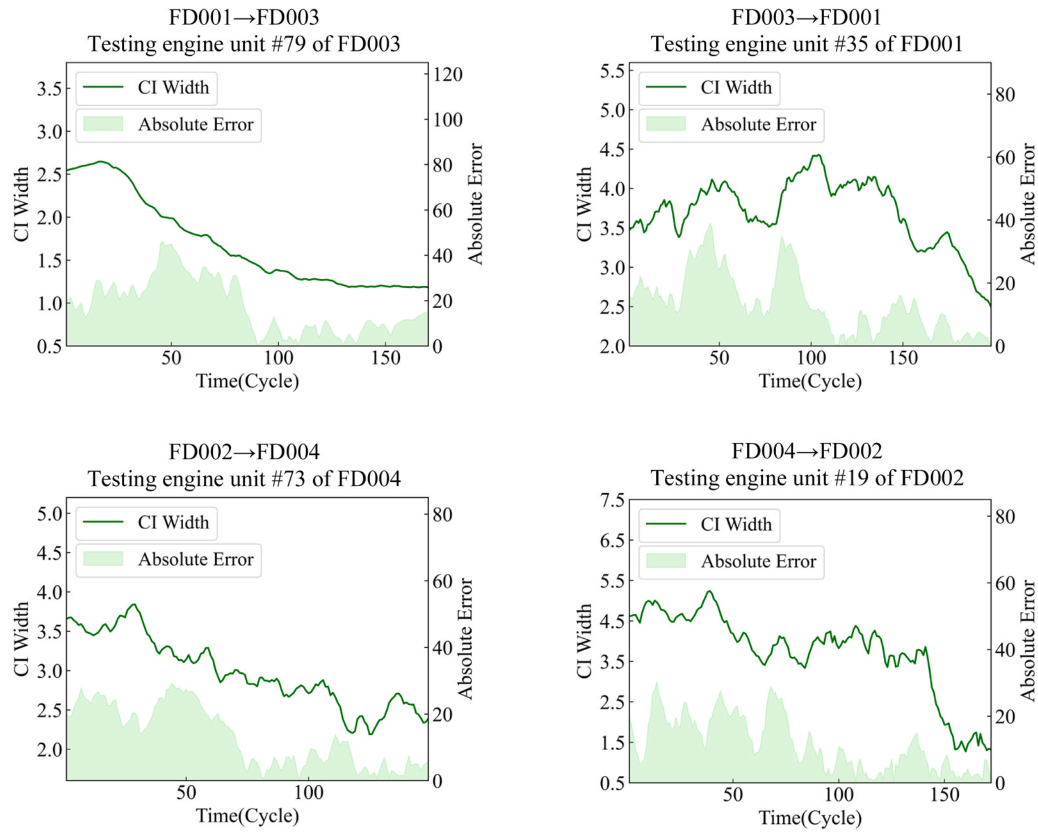

Fig. 14. Absolute error and confidence interval (CI) width.

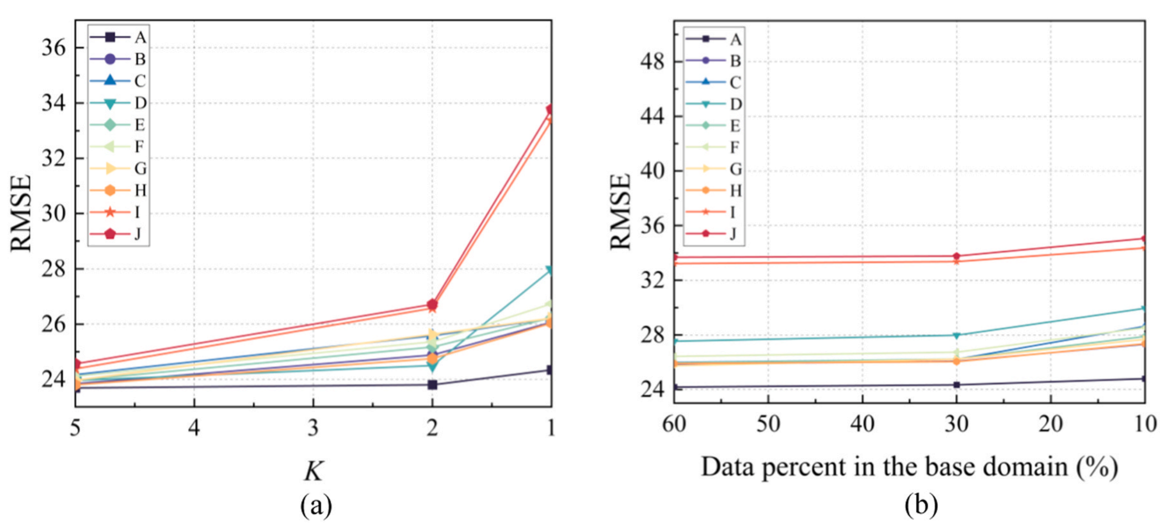

Fig. 15. Experimental results with different amounts of data. (a) the novel domain. (b) the base domain.

across domains ensures that predictive models remain effective despite shifts in usage patterns and environmental factors.

9. Conclusion

In this paper, a novel cross-domain few-shot RUL estimation framework is proposed based on an improved MAML. First, a segmentation strategy is adopted for meta-tasks construction to increase the number of meta-tasks, providing more comprehensive degradation information for efficient meta knowledge extraction of meta-training. Meanwhile, the MAML is improved by task embeddings. The improved MAML operates extremely few labeled data on an efficient low-dimensional parameter space to further alleviate overfitting caused by limited labeled data and exploits representation change for better adaptability in cross-domain scenarios. These contributions have substantially improved the RUL estimation performance, as demonstrated by notable enhancements in key performance metrics throughout our experiments. Compared to the baseline, the proposed framework achieves a 10 % reduction in the RMSE metric and a 35 % reduction in the Score metric, clearly demonstrating the effectiveness of these contributions. In comparison to established prognostic frameworks, the proposed framework yields a 9 % reduction in the RMSE metric and a 31 % reduction in the Score

Table 17
Comparison experiments with the established prognostic frameworks in FD001→FD003, FD003→FD001, FD002→FD004, and FD004→FD002.

Frameworks	FD001→FD003		FD003→FD001		FD002→FD004		FD004→FD002	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
DSSANN (Xu et al., 2022)	67.58	89,165	65.75	59,872	61.16	72,529	52.37	48,550
DCRN (Shang et al., 2024)	61.40	42,221	58.33	28,736	46.19	18,165	40.86	19,396
LSTM-DANN (Costa et al., 2020)	42.15	27,318	28.86	7179	36.41	29,102	25.67	13,091
LSTM-ADDA (Duan et al., 2022)	37.52	15,100	29.12	9443	36.78	21,319	28.66	16,752
LSTM-FT (Zhang et al., 2018)	31.74	9884	26.68	2943	30.40	21,165	25.56	10,751
MAML (Mo et al., 2023)	28.11	3658	26.10	2448	29.27	9327	25.03	3960
MAML w/ task embeddings	**24.34**	**1829**	**24.24**	**1605**	**26.96**	**6944**	**23.24**	**2651**
imp	13.44 %	49.99 %	7.10 %	34.43 %	7.90 %	25.56 %	7.13 %	33.05 %

Table 18
Comparison experiments with the established prognostic frameworks in FD001→FD002, FD002→FD001, FD003→FD004, and FD004→FD003.

Frameworks	FD001→FD002		FD002→FD001		FD003→FD004		FD004→FD003	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
DSSANN (Xu et al., 2022)	69.22	241,073	55.88	23,769	70.21	204,064	49.87	15,259
DCRN (Shang et al., 2024)	63.47	145,054	42.34	6618	62.41	104,805	40.63	8511
LSTM-DANN (Costa et al., 2020)	45.76	76,521	33.23	9178	44.82	89,734	33.95	6731
LSTM-ADDA (Duan et al., 2022)	46.81	41,399	37.10	17,374	42.34	20,392	35.35	11,956
LSTM-FT (Zhang et al., 2018)	38.98	17,983	35.34	7071	40.55	18,177	38.09	7592
MAML (Mo et al., 2023)	36.23	10,801	32.73	5664	36.86	11,665	33.12	5356
MAML w/ task embeddings	**32.06**	**7678**	**30.89**	**4036**	**32.37**	**7855**	**31.70**	**4145**
imp	11.51 %	28.91 %	5.62 %	28.74 %	12.18 %	32.66 %	4.29 %	22.61 %

Table 19
Comparison experiments with the established prognostic frameworks in FD002→FD003, FD003→FD002, FD001→FD004, and FD004→FD001.

Frameworks	FD002→FD003		FD003→FD002		FD001→FD004		FD004→FD001	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
DSSANN (Xu et al., 2022)	58.22	29,816	66.82	191,638	72.46	280,460	50.85	14,034
DCRN (Shang et al., 2024)	43.56	8811	59.39	96,012	66.72	161,743	41.79	9055
LSTM-DANN (Costa et al., 2020)	42.19	15,516	43.90	26,545	43.64	55,382	35.73	8724
LSTM-ADDA (Duan et al., 2022)	39.50	7916	41.76	23,570	44.69	33,271	34.83	47,523
LSTM-FT (Zhang et al., 2018)	41.03	9147	41.51	21,862	41.36	25,721	37.69	8516
MAML (Mo et al., 2023)	37.32	6695	37.28	12,740	37.91	13,244	34.28	5872
MAML w/ task embeddings	**33.21**	**4534**	**33.16**	**8965**	**33.43**	**9366**	**32.71**	**4377**
imp	11.01 %	32.28 %	11.05 %	29.63 %	11.82 %	29.28 %	4.58 %	25.46 %

metric, further highlighting the advantages of the proposed framework over others. Additionally, in most cross-domain scenarios, the CV value of the RMSE metric for the proposed framework remains below 2 %, and the CV value of the Score metric stays below 20 %, underscoring the robustness of the framework and affirming the reliability of the experimental results. The improved estimation performance provides a robust basis for subsequent maintenance and management, particularly in scenarios with extremely limited data, thus effectively ensuring the safety, reliability, and economy of the mechanical systems. However, certain limitations remain, such as the framework's reliance on single-modal data, its offline learning nature, and the challenges of generalizing to more complex cross-machine scenarios.

In future research, several directions can be explored to address these limitations and further enhance the framework's applicability. First, integrating multi-modal data (e.g., vibration, thermal, acoustic, and visual signals) into the framework could provide richer degradation information, potentially improving estimation accuracy and robustness. Second, extending the framework to support online learning would enable real-time RUL estimation and dynamic model updates, addressing the current offline learning limitation. Finally, some more complicated estimation scenarios, such as cross machine scenarios, can be taken into consideration. The integration of domain adaptation techniques could be explored to handle more pronounced domain shifts. These future directions aim to build on the current framework's strengths while addressing its limitations, ultimately advancing the field of PHM.

CRediT authorship contribution statement

Xinyu Shang: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Jie Shang:** Writing – review & editing, Supervision, Methodology. **Mingyu Li:** Writing – review & editing, Supervision, Methodology. **Haobo Qiu:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Liang Gao:** Resources, Funding acquisition. **Danyang Xu:** Writing – review & editing, Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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